

Thrombotic Microangiopathies, Including Hemolytic Uremic Syndrome

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DEFINITIONS

Thrombotic microangiopathy (TMA) is a lesion of arteriolar and capillary vessel wall thickening with intraluminal platelet thrombosis and a partial or complete obstruction of the vessel lumina. Laboratory features of thrombocytopenia and microangiopathic hemolytic anemia are almost invariably present. Depending on whether kidney or brain lesions prevail, two pathologically indistinguishable but clinically different entities have been described: hemolytic uremic syndrome (HUS) and thrombotic thrombocytopenic purpura (TTP). Because HUS can involve extrarenal manifestations and TTP may be associated with severe kidney disease, the two can be difficult to distinguish on clinical grounds.¹ Compared with HUS, TTP is associated with more severe thrombocytopenia and less severe acute kidney injury (AKI), but changes in platelet count and kidney function largely overlap in HUS and TTP. Composite scores such as the PLASMIC score have been proposed to identify TTP patients²; however, there are no cut-off values that definitely discriminate the two syndromes. The identified pathophysiologic mechanisms have allowed for the differentiation of the two syndromes on a pathogenetic basis and have paved the way to specific diagnosis and treatment (Table 30.1 and Fig. 30.1).

The term HUS was introduced in 1955 by Gasser and coworkers for an acute fatal syndrome characterized by hemolytic anemia, thrombocytopenia, and severe AKI. HUS occurs most frequently in children younger than 5 years (incidence 5–6 per 100,000 children per year compared with an overall incidence of 0.5–1 per 100,000 per year). About 90% of cases are associated with infection by Shiga-like toxin (Stx) producing *Escherichia coli* (STEC) and, less frequently, with sepsis by *Shigella dysenteriae*. STEC-HUS in 90% of cases is preceded by diarrhea, often bloody. STEC-HUS occurs primarily in children, except in epidemics, when it may occur in patients with a wider range of ages. In 2011 Northern Germany experienced one of the largest STEC-HUS outbreaks ever reported, with 3816 cases of *E. coli* O104:H4 infection, of whom 845 developed STEC-HUS. Almost 90% of affected patients were adults, in particular women.³ Approximately 10% of HUS cases are not caused by Stx-producing bacteria and have been classified as atypical.⁴

Atypical HUS has an annual incidence of 0.5 to 2 per million per year, can occur at any age, and is a very severe disease. Before the introduction of complement inhibition therapy, 50% of patients with atypical HUS progressed to end-stage kidney disease (ESKD), 30% had neurologic symptoms, and 25% died in the acute phase.⁵ Pulmonary, cardiac, and gastrointestinal (GI) manifestations also can occur.

TTP was first described in 1925 by Moschowitz in a teenage girl with fever, hemolytic anemia, bleeding, AKI, neurologic involvement, and a fulminant clinical course. Pathologic changes were characterized by widespread hyaline thrombosis of small vessels. TTP has an annual incidence between 1.5 and 6 cases per million in Europe and the United

States.⁵ TTP is more common in adults aged 40 years on average but can affect any age group and classically presents with the pentad of thrombocytopenia, microangiopathic hemolytic anemia, fever, and neurologic and kidney dysfunction. Central nervous system (CNS) involvement is seen in over 90% of patients and is characterized by thromboocclusive disease of the gray matter. Clinical features include headache, cranial nerve palsies, confusion, stupor, and coma. Up to half of patients with neurologic involvement may be left with sequelae. Up to 25% of patients exhibit creatinine clearance of less than 40 mL/min during long-term follow-up. Cardiac involvement is common in TTP, ranging from elevation of biomarkers to heart failure, infarction, and sudden cardiac death.⁶

LABORATORY FINDINGS

Thrombocytopenia and microangiopathic hemolytic anemia are almost invariably present in TMA patients.⁴ Thrombocytopenia occurs because of platelet consumption by platelet-rich thrombi in the microcirculation evidenced by giant platelets in the peripheral smear, reduced platelet survival time, or both. Thrombocytopenia may be severe but is usually less so in patients with predominant kidney involvement. In children with STEC-HUS, the duration of thrombocytopenia is variable and does not correlate with the course of kidney disease. Microangiopathic hemolysis is likely caused by the passage of blood through the damaged capillaries and arterioles occluded by thrombi, but other explanations have been suggested.⁷ Hemoglobin levels are low (<10 g/dL shifteterin >90% of patients). The peripheral smear reveals reticulocytosis, increased schistocyte numbers, with polychromasia and often nucleated red blood cells (RBCs). Detection of fragmented erythrocytes is crucial to confirm the microangiopathic hemolytic anemia—provided that valvular heart disease and other arterial abnormalities causing erythrocyte fragmentation are excluded. Other indicators of intravascular hemolysis include elevated lactate dehydrogenase (LDH), increased indirect bilirubin, and low haptoglobin levels.⁴ The Coombs test is negative. Serum C3 levels may be low during the acute phase of STEC and atypical HUS and in TTP. Moderate leukocytosis may be present. Bone marrow biopsy shows erythroid hyperplasia and increased megakaryocytes. The normal prothrombin time, partial thromboplastin time, fibrinogen level, and coagulation factors differentiate TMA from disseminated intravascular coagulation. Mild fibrinolysis with minimal elevation in fibrin degradation products may be observed.

Kidney involvement is present in all patients with HUS (by definition) and in about 25% of patients with TTP^{1,8} and manifests with hematuria, proteinuria, occasionally in the nephrotic range, and kidney failure of variable severity. Hemoglobinuria is a sensitive marker of ongoing hemolysis and monitoring for it may help early detection of disease recurrence.

TABLE 30.1 Classification of Hemolytic Uremic Syndrome and Thrombotic Thrombocytopenic Purpura According to Underlying Etiology

Clinical Presentation	Etiology
Hemolytic Uremic Syndrome	
Stx associated	Infections by Shiga toxin–producing bacteria
Atypical hemolytic uremic syndrome	Complement gene abnormalities: 60% overall <i>CFH</i> , 20%–30%; <i>CFI</i> , 4–8%; <i>C3</i> , 8%–10%; <i>MCP</i> , 8%–10%; <i>THBD</i> , 3%–4%; <i>CFB</i> , 1%–2% <i>CFH/CFHR</i> hybrid genes: 3%–5% <i>DGKE</i> : 10% of patients <1 year old anti- <i>CFH</i> antibodies: 5%–10%
Secondary Hemolytic Uremic Syndrome	
Streptococcus pneumoniae	Desialylation of cells by neuroaminidase Complement gene abnormalities: 20%–30%
Pregnancy associated	Complement gene abnormalities: 50%–60%
Malignant hypertension	Complement gene abnormalities: 35%–65%
Transplantation (de novo HUS)	Calcineurin inhibitors, ischemia, immune response Complement gene abnormalities: 30%
BM/HSC transplantation	Immunosuppression, chemotherapy, GVHD, radiation, infections Complement gene abnormalities: 20%–60%
Autoimmune and systemic diseases	Autoimmune endothelial damage, secondary complement activation Complement gene abnormalities: around 10%
Drugs	Direct toxicity to endothelial cells Complement gene abnormalities rare: <1%
Infections	Endothelial injury, complement activation Complement gene abnormalities: variable %
Malignancies	Unknown
Cobalamin C deficiency	<i>MMACHC</i> mutations
Thrombotic Thrombocytopenic Purpura	
Genetic	Homozygous or compound heterozygous mutations in <i>ADAMTS13</i> gene
Acquired	Anti- <i>ADAMTS13</i> autoantibodies

BM/HSC, Bone marrow/hematopoietic stem cell; *CFH*, complement factor H; *GVHD*, graft-versus-host disease; *HUS*, hemolytic uremic syndrome.

PATHOLOGY

TMA manifests as widening of the subendothelial space and microvascular thrombosis. Electron microscopy (EM) shows characteristic lesions of swelling and detachment of the endothelial cells from the basement membrane and the accumulation of fluffy material in the subendothelium, intraluminal platelet thrombi, and partial or complete obstruction of vessel lumina (Figs. 30.2 and 30.3).⁹ These lesions are similar to scleroderma, malignant nephrosclerosis, chronic transplant rejection, and calcineurin inhibitor nephrotoxicity. Microthrombi are present primarily in the kidneys in HUS and in brain in TTP. In very young children and in those with STEC-HUS, the glomerular injury predominates (Fig. 30.4).⁹

Thrombi and leukocyte infiltration are common in the early phases and usually resolve after 2 to 3 weeks. Patchy cortical necrosis may be present in severe cases; crescent formation is uncommon. In idiopathic and familial forms and in adults, the injury mostly involves arteries and arterioles, with thrombosis and intimal thickening (Fig. 30.5; see also Fig. 30.3), secondary glomerular ischemia, and retraction of the glomerular tuft (Fig. 30.6). Focal segmental glomerulosclerosis may follow acute HUS and is usually seen in children with long-lasting hypertension and progressive chronic kidney disease (CKD).

The typical pathologic changes of TTP are widespread occlusive capillary and arteriolar thrombi most commonly detected in kidneys, pancreas, heart, adrenals, and brain. Compared with HUS, pathologic changes of TTP are more extensively distributed.

MECHANISMS, CLINICAL FEATURES, AND MANAGEMENT OF SPECIFIC FORMS OF THROMBOTIC MICROANGIOPATHY

Shiga Toxin–Producing *Escherichia coli*–Associated Hemolytic Uremic Syndrome

Mechanisms

STEC-HUS may follow GI infections by certain strains of *E. coli* or *S. dysenteriae* that produce Stxs 1 and 2 (see Fig. 30.1 and Table 30.1).¹⁰ Most patients present with bloody diarrhea that may still be present or resolved at the time of presentation of HUS. Multiple strains of *E. coli* isolated from human cases with diarrhea produce Stxs (STEC). Before 2007, more than 70% of US cases of STEC-HUS were ascribed to the serotype O157:H7. However, starting from 2005, the reports of cases caused by non-O157 STEC increased and during 2010 to 2017, more than half of cases have been associated with infections by about 200 non-O157 STEC, with the predominance of serogroups O26, O103, O111, O121, O145, O45, O91, O146, and O113.^{11,12} The natural reservoir of STEC is the GI tract of ruminants (mainly cattle). Most STEC-infected cattle remain free of disease because they lack vascular Stx receptors. Undercooked ground beef, meat patties, raw vegetables, fruit, milk, and recreational or drinking water contaminated by ruminants' excreta have been implicated in the transmission of STEC.

E. coli O157:H7 and other STECs have been responsible for multiple outbreaks worldwide.¹⁰ The very large HUS outbreak in Germany in 2011 was caused by ingestion of fenugreek sprouts contaminated by an unusual STEC strain, O104:H4 related to fecal contamination of the seeds in Egypt.³ Because no O104:H4 strains were detected in cattle feces collected in the outbreak area, infected ruminants are unlikely to be the cause of the O104:H4 STEC outbreak.

After food or water is ingested, STEC colonize the intestinal mucosa through an attaching/effacing lesion, characterized by the disruption of microvilli and attachment of the bacteria to the enterocytes.¹³ This promotes diarrhea and intestinal inflammation. Released Stxs damage blood vessels in the colon. Very low levels of Stxs occur in the sera of HUS patients; however, Stxs can bind to blood cells or blood-derived microvesicles, which carry Stx from the intestine to target organs. Erythrocytes, platelets, and monocytes bind Stxs via specific globotriaosylceramide (Gb3) receptors, whereas human neutrophils interact with Stxs through the toll-like receptor 4.¹³

In the kidney, Stxs bind mainly to specific Gb3 receptors on glomerular endothelial cells but also to podocytes, mesangial cells, and proximal tubular cells. Bound toxin is internalized and inhibits protein synthesis, causing cell death. Treatment of endothelial cells with sublethal doses of Stx increased mRNA and protein expression of chemokines and cell adhesion molecules, favoring leukocyte-dependent

Investigation, Diagnosis, and Management of Thrombotic Microangiopathies

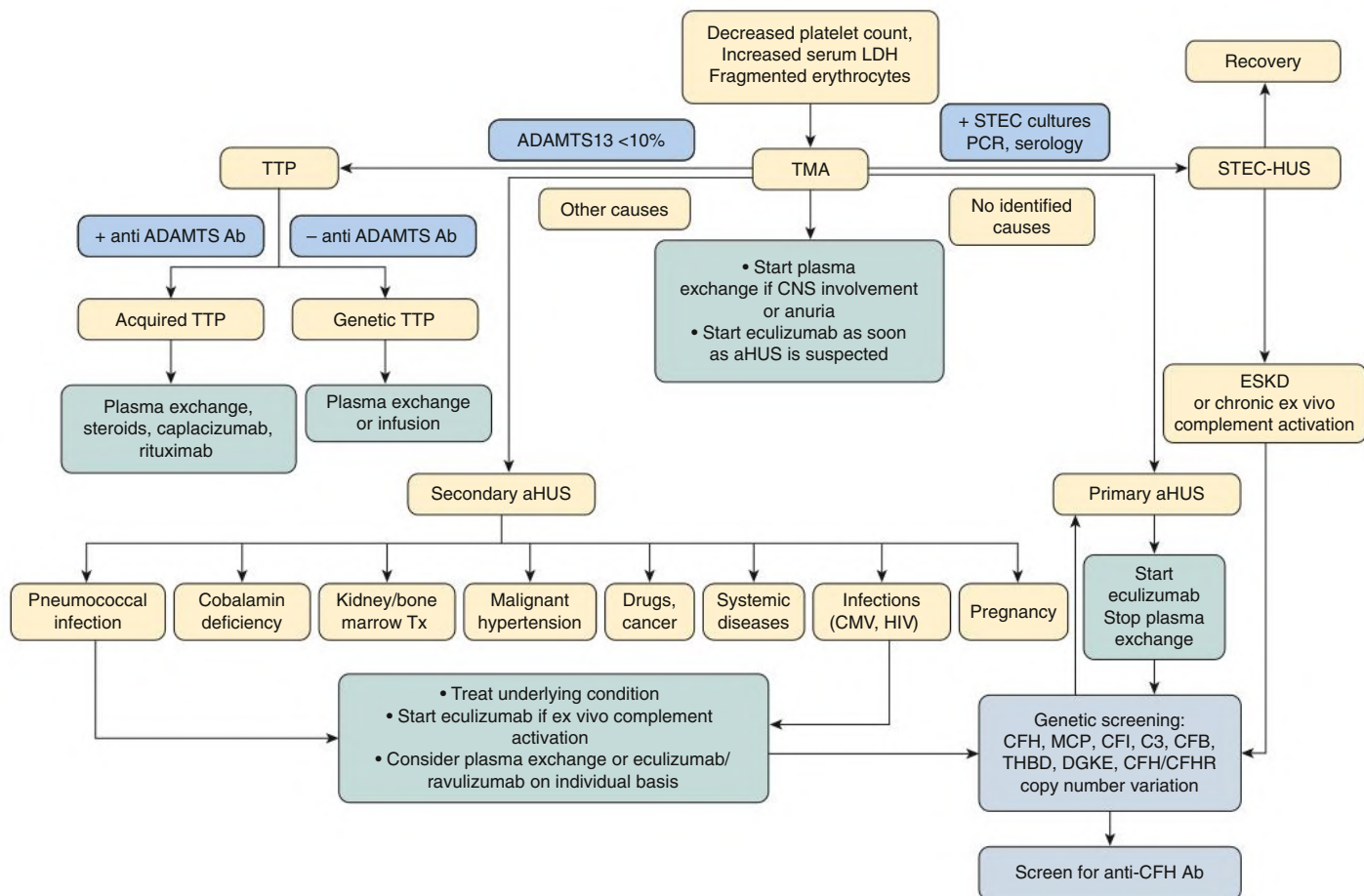


Fig. 30.1 Investigation, Diagnosis, and Management of Thrombotic Microangiopathies. We recommend empirical treatment with eculizumab in particular if there is evidence for in vivo complement activation (and ideally ex vivo complement activation if such diagnostic testing is available). This latter testing involves evaluating for serum-induced complement deposition on cultured endothelial cells and is done only in a few specialized centers. *Ab*, Antibody; *aHUS*, atypical hemolytic uremic syndrome; *CFB*, complement regulatory factor B; *CFH*, complement regulatory factor H; *CFI*, complement regulatory factor I; *CMV*, cytomegalovirus; *CNS*, central nervous system; *DGKE*, diacylglycerol kinase epsilon; *ESKD*, end-stage kidney disease; *HIV*, human immunodeficiency virus; *LDH*, lactate dehydrogenase; *MCP*, membrane cofactor protein (CD46); *MLPA*, multiplex ligation-dependent probe amplification; *STEC-HUS*, Shiga toxin-producing *Escherichia coli*-associated hemolytic uremic syndrome; *THBD*, thrombomodulin; *TMA*, thrombotic microangiopathy; *TTP*, thrombotic thrombocytopenic purpura; *Tx*, transplantation.

inflammation and inducing loss of thromboresistance in endothelial cells.

Complement activation at the renal endothelial level may contribute to microangiopathic lesions in STEC-HUS. Low serum levels of C3 and increased plasma levels of C3b, C3c, and C3d occur in children with STEC-HUS. High levels of complement activation products Bb and C5b-9 indicate complement activation via the alternative pathway.¹³ In addition, C3 and C9 deposit on the surface of blood-derived microparticles from STEC-HUS patients. Stxs might directly contribute to complement activation on endothelial cells, which contributed to loss of thromboresistance.¹⁴ Stx2 also caused the formation of platelet-leukocyte aggregates with surface-bound C3 and C9. After exposure to Stx/lipopolysaccharide (LPS), factor B–deficient mice that cannot activate the alternative pathway of complement exhibited less thrombocytopenia and were protected against glomerular abnormalities

and kidney function impairment.¹⁴ Another glomerular target for Stx-induced complement activation is the podocyte, as documented by podocyte depletion—a potential cause of proteinuria—in Stx/LPS mice, which was limited by treatment with a C3a receptor antagonist.¹⁴

Tubular epithelial and mesangial cells are also susceptible to the cytotoxic effects of Stxs. Stxs inhibit water absorption across human renal tubular epithelial cell monolayers, which may contribute to the early events underlying kidney dysfunction in STEC-HUS.

Diagnosis

Diagnosis depends on the detection of *E. coli* O157:H7 and other STECs and their products in stool cultures (see Fig. 30.1). When infection with STEC is suspected, physicians should ensure that stool specimens are collected promptly and specifically cultured for STEC.¹⁰ Unlike most other *E. coli*, serotype O157:H7 does not

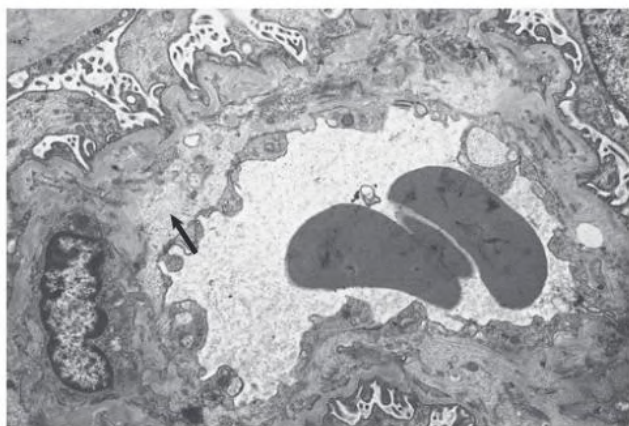


Fig. 30.2 Electron Micrograph of a Glomerular Capillary in Hemolytic-Uremic Syndrome. The endothelium is detached from the glomerular basement membrane (GBM); the subendothelial space is widened and occupied by electron-lucent fluffy material and cell debris (arrow). Beneath the endothelium is a thin layer of newly formed GBM.

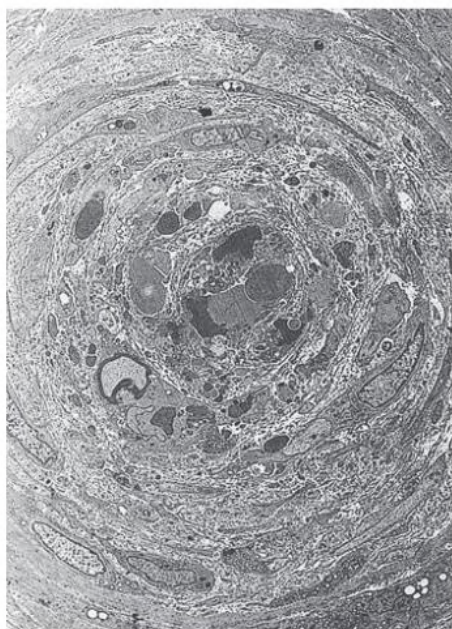


Fig. 30.3 Electron Micrograph of a Renal Arteriole in Hemolytic Uremic Syndrome. The vascular lumen is completely occluded by thrombi. There is marked intimal edema with consequent separation of myointimal cells.

ferment sorbitol rapidly and thus forms colorless colonies on sorbitol containing MacConkey agar (SMAC). The use of SMAC provides a simple, inexpensive, and generally reliable method of screening stools for *E. coli* O157. The polymerase chain reaction to detect *Stx*-encoding genes using DNA directly isolated from stool specimens has substantially improved the identification of non-O157:H7 STEC strains. Convalescent-phase serum samples can be assayed for antibodies to O157 or other specific strain-derived LPS; however, false-positive results may occur because of antibodies preformed during antecedent STEC exposure.¹⁰

Clinical Course

After exposure to STEC, 38% to 61% of individuals develop hemorrhagic colitis and 3% to 9% (in sporadic infections) to 20% (in epidemic forms) progress to overt HUS (Fig. 30.7).¹⁰

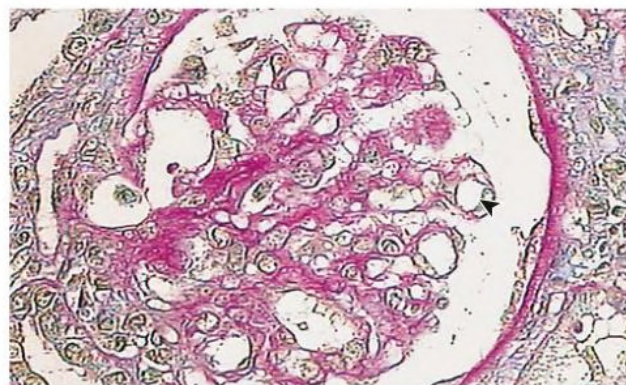


Fig. 30.4 Glomerulus From a Patient With Shiga Toxin–Producing *Escherichia coli*-Associated Hemolytic Uremic Syndrome. A marked thickening of the glomerular capillary wall occurs with many double contours (arrowhead).

STEC-induced hemorrhagic colitis without HUS is self-limiting and is not associated with a long-term risk for CKD. STEC-HUS is characterized by prodromal diarrhea followed by AKI. The average interval between *E. coli* exposure and illness is 3 days. Illness typically begins with abdominal cramps and nonbloody diarrhea; diarrhea may become hemorrhagic in 70% of cases, usually within 1 or 2 days.¹⁰ Vomiting occurs in 30% to 60% of cases and fever in 30%. HUS is usually diagnosed 6 to 10 days after the onset of diarrhea. Of those who develop HUS, 70% HUS require RBC transfusions, 40% to 50% need dialysis for an average duration of 10 days, and the remainder have milder kidney involvement.^{10,15} About 25% of patients with STEC-HUS have neurologic involvement, including lethargy, apnea, cortical blindness, hemiparesis, stroke, seizures, and coma. GI complications include bowel ischemia/necrosis and perforation. Rare complications include pancreatitis, diabetes mellitus, myocardial ischemia because of TMA in coronary arteries and microvasculature, and pleural and pericardial effusions¹⁶; 1% to 4% of patients die during the acute phase.

About 70% of childhood cases of STEC-HUS fully recover from the acute disease. Those who do not fully recover may have persistent proteinuria (15%–30% of patients), hypertension (5%–15%), CKD (9%–18%), or ESKD (3%).¹⁵

Compared with previous STEC epidemics, during the STEC O104:H4 German outbreak, there was a higher incidence of dialysis-dependent AKI (20% vs. 6%) and death (6% vs. 1%).³ Nearly half of the patients presented with neurologic symptoms and 20% had seizures. The severe clinical phenotype was explained by the lack of previous immunity to this novel STEC strain and its exceptional virulence.³

Therapy

Typical pediatric STEC-HUS treatment relies on supportive management of anemia, kidney failure, hypertension, and electrolyte and water imbalance (see Fig. 30.1). Intravenous (IV) isotonic volume expansion as soon as an *E. coli* O157:H7 infection is suspected, even before culture results are available, may limit the severity of AKI and the need for kidney replacement therapy.¹⁷ Up to 80% of patients need RBC transfusions. Heparin and antithrombotic agents increase the risk for bleeding and should be avoided. Patients with severe STEC-HUS require careful monitoring, including urine output, weight, volume status, cardiovascular/respiratory function, and early signs of CNS or other organ involvement. Bowel rest (i.e., complete parenteral nutrition) is important for the enterohemorrhagic colitis associated with STEC-HUS. Antimotility agents should be avoided because they may prolong the persistence of *E. coli* in the intestinal lumen and therefore increase patient exposure to its toxin.

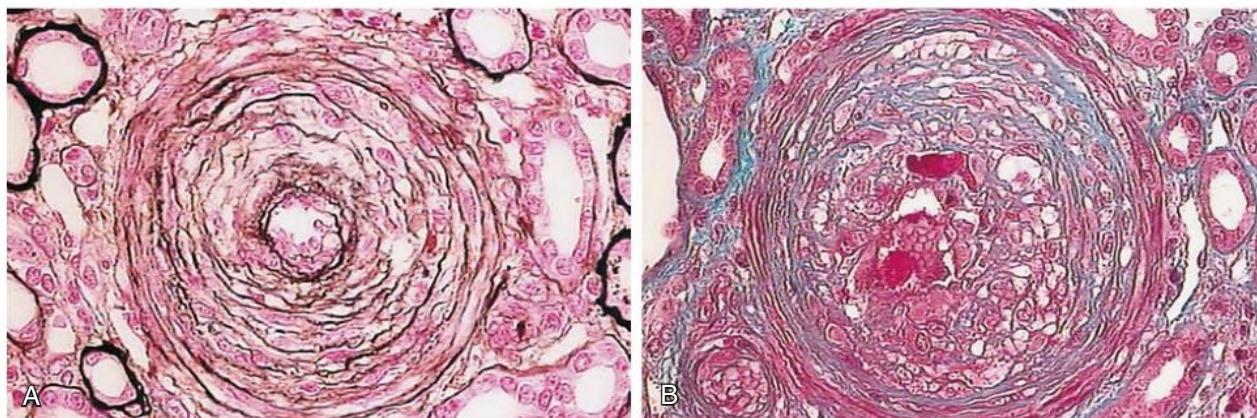


Fig. 30.5 Interlobular Artery in a Case of Hemolytic-Uremic Syndrome With Severe Vascular Involvement. (A) The vascular lumen is almost completely occluded. Changes include myointimal proliferation and reduplication of the lamina elastica. (B) Thrombotic material and erythrocytes can be seen in the lumen and within the vascular wall.

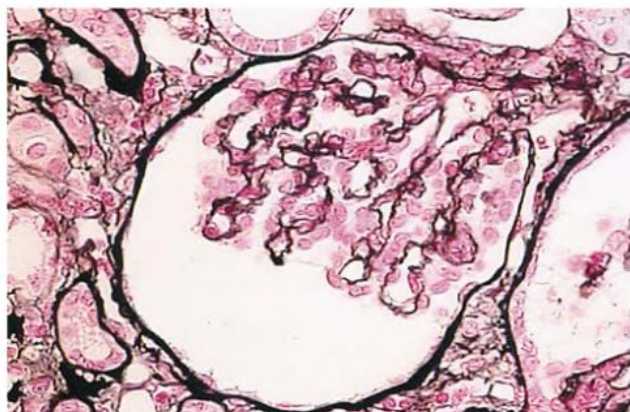


Fig. 30.6 Glomerulus From a Patient With Atypical Hemolytic Uremic Syndrome With Predominant Vascular Involvement. Severe ischemic changes have occurred. Note the shrinkage of the glomerular tuft and marked thickening and wrinkling of the capillary wall.

Antibiotics should be restricted to the rare patients presenting with bacteremia because, at least in children with gastroenteritis, antibiotic treatment may increase the risk for HUS 17-fold, possibly related to acute release of large amounts of preformed toxin from lysed bacteria.¹⁸ Alternatively, antibiotics might give *E. coli* O157:H7 a selective growth advantage. Moreover, several antimicrobial drugs, particularly the quinolones, trimethoprim, and furazolidone, are potent inducers of the expression of the Stx gene. Azithromycin may be an exception because its use reduced the duration of bacterial shedding in adults in the German O104:H4 epidemic.¹⁹ In contrast to STEC-HUS, hemorrhagic colitis and HUS caused by *S. dysenteriae* bacteremia and sepsis should be treated with antibiotics because treatment shortens the duration of diarrhea, decreases the incidence of complications, and reduces the risk for transmission.

Careful blood pressure control and renin-angiotensin system blockade may be beneficial in the long term for patients who develop CKD after an episode of STEC-HUS.

Newer potential treatments for STEC-HUS, including Stx-binding agents and Stx-neutralizing antibodies, have been abandoned because of lack of efficacy. The only published data on the Stx-binding agent SYNORB Pk failed to show benefits.¹⁷ Other potential therapeutics under development are designed to limit Stx receptor expression or to prevent toxin binding, trafficking, or activity within the cells.²⁰

Timing of Events That May Follow Exposure to Shiga Toxin-Producing *E. coli*

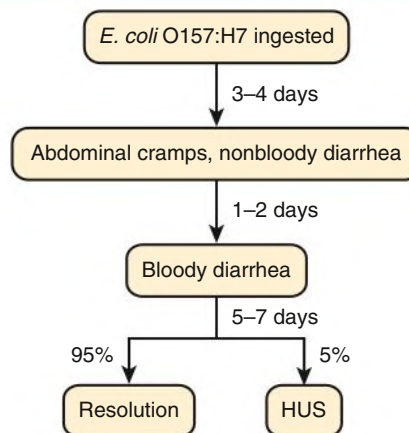


Fig. 30.7 Timing of events that may follow exposure to Shiga toxin-producing *E. coli*. HUS, Hemolytic uremic syndrome.

The efficacy of specific treatments in adult patients is difficult to evaluate because most information is derived from uncontrolled series that may include atypical HUS cases. No prospective randomized trials demonstrate whether plasma infusion or exchange are superior to supportive treatment alone (Table 30.2). However, comparative analyses of two large series of patients treated²¹ or not treated²² with plasma suggest that plasma therapy may dramatically decrease overall mortality of STEC O157:H7-associated HUS. Plasma exchange should therefore be considered in adult patients with severe AKI and CNS involvement.

Kidney transplantation is safe for children who develop ESKD. Recurrence rates range from 0% to 10%, and graft survival at 10 years is better than in children with other causes of ESKD.

Evidence that uncontrolled complement activation may contribute to microangiopathic lesions of STEC-HUS¹⁴ led to complement inhibitor therapy with the anti-C5 monoclonal antibody eculizumab in three children with severe STEC-HUS who fully recovered. However, eculizumab therapy plus plasma exchange in the German STEC O104:H4 outbreak did not improve outcomes compared with plasma exchange alone.²³ These data, however, were retrospectively collected

TABLE 30.2 HUS and TTP: Specific Therapies, Dosing, and Efficacy

Therapy	Dosing	Efficacy
Immunosuppressives		
Prednisone	200 mg tapered to 60 mg/day, then 5 mg reduction per wk	Probably effective in addition to plasma exchange in patients with TTP and anti-ADAMTS13 autoantibodies or in aHUS with anti-factor H autoantibodies and in forms associated with autoimmune diseases. Lack of evidence from controlled trials in immune-mediated HUS or TTP.
Prednisolone	200 mg, tapered to 60 mg/day, then 5 mg reduction per wk	
Immunoglobulins	400 mg/kg/day	
CD20 cell depleting antibody (rituximab)	375 mg/m ² /wk up to CD20 and/or CD19 cell depletion	Effective in treatment or prevention of TTP associated with immune-mediated ADAMTS13 deficiency resistant to or relapsing after immunosuppressive therapy.
Fresh frozen plasma		
Exchange	1–2 plasma volumes/day	First-line therapy in TTP. Unproven efficacy in childhood STEC-HUS. Possibly effective in aHUS if eculizumab is not available.
Infusion	20–30 mL/kg followed by 10–20 mL/kg/day	To be considered if plasma exchange is not available.
Cryosupernatant	See plasma infusion/exchanges	To replace whole plasma in case of resistance to plasma infusion or sensitization.
Solvent-detergent–treated plasma	See plasma infusion/exchanges	To limit the risk for infections.
Caplacizumab	10–11 mg IV loading dose, then 10–11 mg SC daily combined with plasma exchange and immunosuppression	Reported efficacy in acquired TTP.
Liver-kidney transplant	Perioperative plasma infusion/exchange and eculizumab	To prevent CFH-associated HUS recurrence posttransplant; about 20% mortality risk.
Complement inhibition (eculizumab)	900 mg/wk for the first 4 wk 1200 mg every 14 days up to 6 mo in adults and children with body weight >40 kg. In smaller children the dose is adjusted on the basis of body weight.	First-line therapy in primary aHUS.

aHUS, Atypical hemolytic uremic syndrome (HUS); CFH, complement factor H; IV, intravenous; SC, subcutaneous; STEC-HUS, Shiga toxin–producing *Escherichia coli*–associated HUS; TTP, thrombotic thrombocytopenic purpura.

and patients given eculizumab were the most severely ill. Two double-blind placebo controlled trials, ECUSTEC in UK (ISRCTN89553116) and ECULISHU in France (NCT02205541), in severe forms of STEC-HUS will be finalized soon.

Atypical Hemolytic Uremic Syndrome

The term atypical HUS (aHUS) has been used to define any form of HUS that is not caused by STEC.⁴ Several other classifications each carry caveats.²⁴ Currently, the term aHUS is used when a genetic or autoimmune abnormality causing complement dysregulation is strongly suspected (see Fig. 30.1 and Table 30.1)^{4,25} and other causes have been excluded. Secondary HUS instead includes a broad group of patients in whom TMA occurs in the context of another condition such as malignant hypertension, autoimmune disease, certain infections, malignancy, transplantation, or drugs. However, this differentiation is not absolute because in more than 50% of patients with primary aHUS and genetic risk factors, a trigger is required for disease to manifest, such as viral and bacterial infections or pregnancy.^{4,25} Conversely, complement gene abnormalities have been identified in subgroups of patients with secondary HUS, indicating the relevance of genetic background for disease susceptibility.^{25,26} Consistently, measurement of circulating complement parameters and ex vivo assays of complement

deposition on cultured endothelial cells with serum and plasma from patients supported a role of complement activation in the etiology of several forms of secondary HUS.²⁷

Primary aHUS^{4,25} is usually sporadic, and fewer than 20% of cases are familial. Although some are in siblings, suggesting autosomal recessive transmission, others occur across two or three generations, indicating an autosomal dominant mode. Incomplete penetrance of the disease in mutation carriers is a common feature that confounds the interpretation of inheritance. Indeed, many sporadic cases of aHUS inherited the genetic defect from an unaffected parent.²⁸

Mechanisms

About 50% of aHUS patients exhibit reduced serum levels of C3 with normal C4.²⁹ C3 consumption is documented by high levels of activated products and C3 deposits in glomeruli and other kidney vessels. The complement pathways—classic, lectin, and alternative pathways (see Fig. 17.6)—are discussed in Chapter 17. The alternative pathway (Fig. 30.8) is initiated spontaneously in plasma by C3 hydrolysis responsible for deposition of a low amount of C3b onto all plasma-exposed surfaces. On bacterial surfaces, C3b leads to phagocytosis by neutrophils and macrophages. Without regulation, a small initiating stimulus is quickly amplified to a self-harming response until complement

Alternative Pathway of Complement Activation

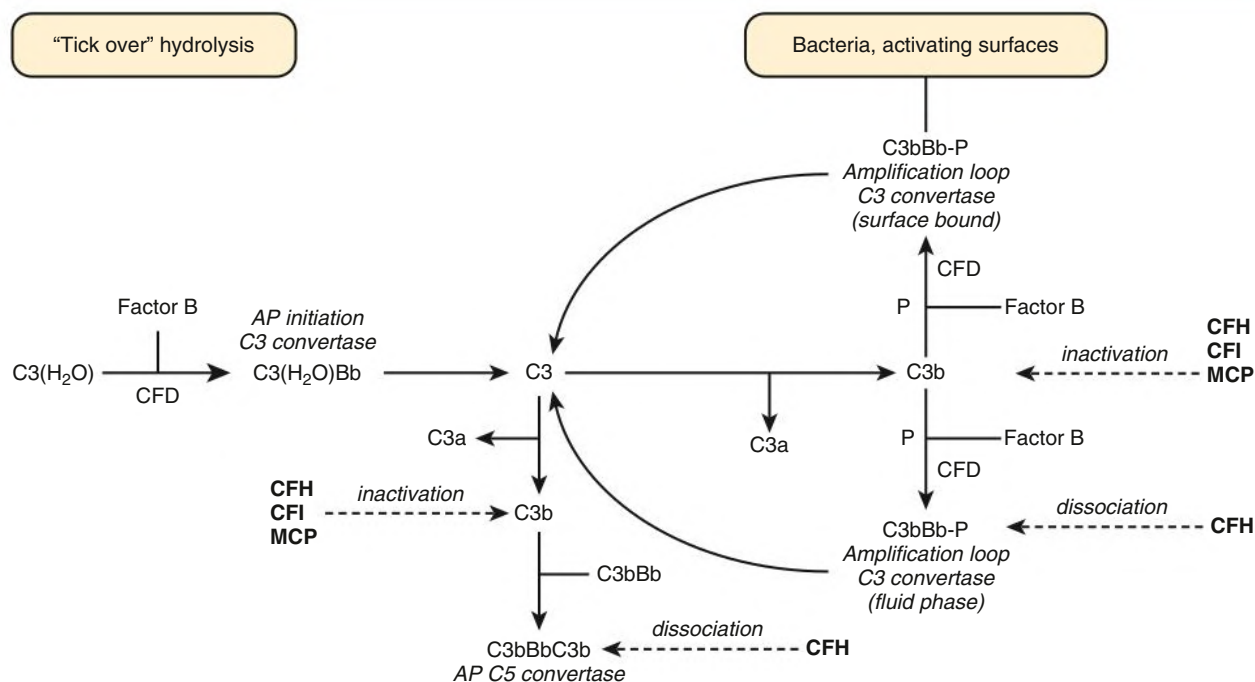


Fig. 30.8 Alternative Pathway of Complement Activation. The alternative pathway (AP) is continuously activated in plasma by low-grade hydrolysis of C3. The latter binds factor B, to form a C3(H₂O)B complex. Factor D (CFD) cleaves factor B to form the AP initiation C3 convertase that cleaves C3 to C3b. The activation is then amplified by the covalent binding of a small amount of C3b to hydroxyl groups on cell surface carbohydrates and proteins of target cells such as bacterial cells. This C3b binds factor B to form the amplification loop C3 convertase C3bBb. C3 convertase enzymes cleave many molecules of C3, resulting in a positive feedback amplification loop. C3b also binds to the C3 convertase, forming the C5 convertase enzyme C3b₂Bb. The AP is highly regulated to prevent nonspecific damage to host cells and limit the deposition of complement to the surface of pathogens. This fine regulation occurs through a number of membrane-anchored and fluid phase regulators. CFH, Complement factor H (acts as cofactor for factor I for C3b cleavage and favors the decay of the C3 convertase of the AP); CFI, complement factor I (degrades C3b and C4b); CR1, complement receptor 1 (has decay accelerating activity and cofactor activity for factor I-mediated cleavage of C3b and C4b); DAF, decay accelerating factor (destabilizes the C3 and C5 convertases of the classic and alternative pathways); MCP, membrane cofactor protein (binds C3b and C4b and has cofactor activity for both the classic and alternative pathways).

components are depleted. On host cells, such a dangerous cascade is controlled by membrane-anchored and fluid-phase regulators (see Fig. 30.8). They both favor the cleavage of C3b to inactive iC3b by the plasma serine protease factor I (CFI, cofactor activity) and dissociate the C3 and C5 convertases (decay acceleration activity). Foreign targets and injured cells that either lack membrane-bound regulators or cannot bind soluble regulators are attacked by complement.

The C3 convertases of the classic/lectin pathways are formed by C2 and C4 fragments, whereas the alternative pathway convertase requires cleavage of C3 only (see Fig. 30.8). Thus, low serum C3 levels in aHUS with normal C4 indicated selective alternative pathway activation.²⁹

Several genetic abnormalities in members of the alternative pathway of complement have been described in aHUS, which account for about 60% of cases (see Table 30.1). Functional studies revealed that aHUS-associated mutations mainly result in complement activation that is restricted on the cell surface—which explains the normal circulating complement profile in about half of patients—and proceeds until the formation of C5b-9.²⁷

Complement factor H. Complement factor H (CFH) regulates the alternative pathway both in the fluid phase and on the cell surface by

acting as a cofactor for CFI and enhancing dissociation of C3 convertase (see Fig. 30.8). Over 120 CFH pathogenetic or likely pathogenetic variants (Database of Complement Gene Variants; www.complement-db.org) have been identified in aHUS patients (mutation frequency about 30%).²⁸ These genetic abnormalities most commonly are associated with normal CFH levels but instead result in dysfunctional protein that cannot bind to and regulate complement on endothelial cells and platelets.³⁰ A high degree of sequence identity between CFH and the genes CFHR1-5 for five factor H-related proteins (CFHR) located in tandem to CFH may predispose to gene conversions and genomic rearrangements.³¹ Hybrid CFH/CFHR1 and CFH/CFHR3 genes, coding abnormal FH proteins in which the carboxy-terminal domains that mediate complement regulation on cell surface are substituted for those of FHR1 or by the entire FHR3, occur in 3% to 5% of patients with aHUS. The resulting hybrid CFH molecules exhibit decreased complement regulatory activity on endothelial surfaces.³¹ Different gene conversion events between the CFH and CFHR1 genes have been reported in unrelated aHUS patients, which convert the FHR-1 C-terminus into that of CFH. The resulting FHR1 mutant strongly competes with CFH for cell surface binding.

Anti-CFH inhibitory antibodies have been reported in 5% to 10% of aHUS patients and around 25% to 50% of pediatric cases.³² These autoantibodies predominantly target the C-terminal end, impairing complement regulation on host cell surfaces. The development of CFH autoantibodies in aHUS has a genetic predisposition, being strongly associated with the homozygous deletion of the *CFHR1* and *CFHR3* genes.³² Complement gene variants identified in a quarter of patients with anti-CFH antibodies may affect disease severity and outcome.

Membrane cofactor protein. Membrane cofactor protein (MCP) is a transmembrane complement regulator widely expressed on all cells apart from erythrocytes. MCP serves as a cofactor for CFI to cleave C3b and C4b (see Fig. 30.8). MCP gene abnormalities account for 8% to 10% of aHUS cases.²⁸ Most are heterozygous (www.complement-db.org). The majority cluster in critical extracellular modules for regulation. Expression on blood leukocytes was reduced for about 75% of mutants. Other MCP mutants have low C3b-binding capability and decreased cofactor activity.²⁸

Complement factor I. CFI is a plasma serine protease that regulates the three complement pathways by cleaving C3b and C4b in the presence of cofactor proteins (see Figs. 17.6 and 30.8). CFI pathogenetic variants affect 4% to 8% of patients. All mutations are heterozygous; 80% cluster in the serine-protease domain. Approximately 50% of mutants are not secreted in blood; however, some mutants are secreted but have impaired proteolytic activity.²⁸

Complement factor B and C3. Gain-of-function mutations can affect genes encoding the alternative pathway C3 convertase components, complement factor B (CFB) and C3. CFB variants are rare (1%–2%). Some CFB mutants have excess C3b affinity and form a hyperactive C3 convertase resistant to dissociation; however, about 60% of mutants have normal activity and are likely unrelated to aHUS pathogenesis.³³

About 9% of aHUS patients carry heterozygous variants in C3, usually with low C3 levels. Most mutations reduce C3b binding to complement regulators, severely impairing its inactivation and result in increased C3 deposition on endothelial cells.

Thrombomodulin. Heterozygous mutations in the gene *THBD* encoding thrombomodulin, an endothelial surface anticoagulant protein that also modulates complement on cell surfaces, have been found in 3% to 4% of patients with aHUS.³⁴ Cells expressing *THBD* variants inactivate C3b less efficiently.³⁴

Determinants of disease penetrance. Incomplete penetrance of aHUS in carriers of complement gene abnormalities ranges from 20% to 50%.²⁸ A further mutation in one of the aforementioned genes occurs in approximately 10% of aHUS patients and increases the risk for developing the disease.³⁵ Common genetic risk variants (single-nucleotide polymorphisms [SNPs] and haplotype blocks) in *CFH*, *MCP*, and *CFHR1* have been shown to act as susceptibility factors for the development of aHUS.³⁵ In a study of 103 aHUS pedigrees, the concurrent presence of two pathogenetic variants and the *CFH* and *MCP* risk haplotypes resulted in 100% penetrance.³⁶

Diacylglycerol kinase ϵ . Homozygous or compound heterozygous mutations in *DGKE*, which encodes the intracellular protein diacylglycerol kinase ϵ , have been recently associated with infantile recessive aHUS.³⁷ Mutation carriers presented with aHUS before 1 year of age (see Table 30.1), with hypertension, hematuria, and severe proteinuria, often in nephrotic range. *DGKE* is expressed in endothelium, platelets, and podocytes; is involved in terminating diacylglycerol signaling; and is not directly linked to complement. However, a subgroup of patients exhibits systemic complement activation, with or without concomitant rare complement gene variants.

Diagnosis of Atypical Hemolytic Uremic Syndrome and Testing for Genetic Mutations

Differential diagnosis of aHUS requires exclusion of infections by STEC or concomitant diseases and conditions or ADAMTS13 deficiency or autoantibodies against ADAMTS13 (see Fig. 30.1).²⁵ Full analysis of disease-associated genes and testing for anti-CFH antibodies is recommended because the nature of the underlying complement defect influences disease progression, the risk of relapses after kidney transplantation, and responses to therapies. Reference laboratories in several countries are equipped for genetic and antibody testing. Although new sequencing techniques have reduced cost and time, a complete evaluation still requires several weeks. Treatment of acute episodes (plasma therapy or eculizumab when available; see later discussion) should be started rapidly after clinical diagnosis, without waiting for results of genetic and anti-CFH antibody tests.

Clinical Course

Of all patients with aHUS, 60% are affected during childhood, irrespective of mutation type.²⁸ Almost all patients with anti-CFH antibodies develop the disease before 16 years of age. Acute episodes manifest with severe hemolytic anemia, thrombocytopenia, and AKI. Extrarenal involvement (CNS or multivisceral) occurs in 20% of cases.^{4,28} Before the introduction of anti-C5 therapy, 50% to 70% of patients with *CFH*, *CF*, *C3*, *CFB*, or *THBD* mutations and 40% of children with anti-CFH autoantibodies lost kidney function, died during the presenting episode, or developed ESKD after relapses.^{4,28} Chronic complement dysregulation may lead to atheroma-like lesions enhancing cardiovascular morbidity and mortality. In MCP-mutation carriers, recurrences were frequent but long-term outcome was good and 80% of patients remained dialysis-free.^{4,28} However, rare patients with MCP mutations had severe disease, immediate ESKD, intractable hypertension, and coma, possibly because of concurrent genetic abnormalities.³⁵

Therapy

Fresh frozen plasma. Plasma therapy (plasma exchange, 1–2 plasma volumes/day; plasma infusion, 20–30 mL/kg/day; see Table 30.2) should be started within 24 hours of diagnosis and then continued until remission or a declaration of nonresponse.⁴ Plasma infusion or exchange provides normal CFH to patients carrying *CFH* mutations.²⁸ Long-term treatment, however, may fail as a result of development of plasma resistance. Heterozygous *CFH* mutation carriers usually have normal levels of CFH, half of which is dysfunctional. The beneficial effect of plasma is strongly dependent on the amount, frequency, and modality of administration, with plasma exchange being superior to plasma infusion by removing the mutant CFH that could antagonize the normal protein. Plasma exchange in association with immunosuppressive therapy (induction with corticosteroids and cyclophosphamide and maintenance with azathioprine or mycophenolate mofetil) is recommended in patients with anti-CFH antibodies to remove the inhibitory antibodies^{25,32} and allowed long-term dialysis-free survival in 60% to 70% of patients.³² Data on the effect of rituximab in such circumstances are scanty and inconsistent: of five patients treated with rituximab alone or with plasma exchange, only two showed disappearance of anti-CFH antibodies.³²

Patients with *CFI* mutations showed a partial response to plasma^{4,28}; 30% to 40% of patients with *CFB* mutations and 50% of those with *C3* mutations responded to plasma infusion or exchange.⁴ Possibly these patients need frequent large-volume plasma exchange to clear the hyper-functional mutant *CFB* and *C3*.⁴ Because MCP is a cell-associated protein, effects of plasma are unlikely in patients with MCP mutations. Indeed, 80% of patients with an MCP mutation underwent spontaneous remission independently of plasma treatment.^{4,28}

Kidney transplantation. Disease recurred in 60% to 80% of transplanted patients with mutations in complement-circulating proteins (CFH, CFI, CFB, and C3), and graft failure occurred in 80% to 90%.³⁸ The risk for post-transplant aHUS recurrence in patients with anti-CFH autoantibodies is not well known.³² Recurrence of HUS was documented in 30% of cases. A reduction in autoantibody levels with plasma exchange, steroids, and/or rituximab enabled successful kidney transplantation in few patients.³²

The lowest incidence of recurrence was observed in patients with MCP and DGKE mutations.^{37,38} MCP and DGKE are highly expressed in the kidney, and a graft that brings normal proteins corrects the defect. However, about 20% of patients with MCP mutations also carry a mutation in another complement gene. Such patients have a worse graft outcome with higher incidence of recurrences than patients with an isolated MCP mutation.³⁵

Testing affected patients for mutations on all disease-associated genes should allow patients and clinicians to make informed decisions regarding listing for transplantation based on risk for recurrence.

Most studies have shown that plasma exchange therapy fails to prevent graft loss in patients with recurrent posttransplant aHUS.³⁸ A preemptive plasma infusion or exchange strategy has been successful in preventing recurrent aHUS in kidney transplant recipients,³⁹ although aHUS recurred in some when plasma therapy was tapered.

Living-related kidney donation is contraindicated given the high risk for recurrence, and it even may be risky to donors because uninephrectomy may precipitate aHUS if they are complement gene mutation carriers.

Complement inhibitors. The humanized anti-C5 monoclonal antibody eculizumab induces remission of acute episodes of aHUS and maintains long-term remission, both in native kidneys and in the kidney grafts, as documented in two prospective clinical trials of primarily adult patients with plasma-dependent or plasma-resistant aHUS⁴⁰ and in a pediatric trial in 22 children, of whom 12 had no prior plasma treatment. Eculizumab is now widely used as a first-line therapy for aHUS, provided that other causes of TMA are excluded (see Fig. 30.1 and Table 30.2).²⁵

DGKE-mediated aHUS is generally nonresponsive to eculizumab, with the exception of patients who also present with concomitant complement abnormalities.

Eculizumab prophylaxis is used to prevent aHUS relapses after kidney transplantation. In a study of 126 kidney transplants, eculizumab prophylaxis was independently associated with longer graft survival.⁴¹ However, this was a retrospective study and controlled prospective studies are required to evaluate the advantage, if any, of eculizumab prophylaxis versus eculizumab treatment at the time of overt aHUS recurrence in kidney transplant patients.³⁹

Ravulizumab, an anti-C5 antibody engineered from eculizumab to prolong half-life, has been recently approved for the treatment of aHUS. In clinical trials, rates of remission were similar to those observed in eculizumab trials; however, fewer patients in the ravulizumab study were able to stop dialysis, possibly because of differences in patient cohorts.⁴²

The main concern with eculizumab and ravulizumab is increased susceptibility to infection with encapsulated organisms, particularly *Neisseria* infections. For this reason, patients must receive meningococcal vaccination. Patients should receive vaccination against meningococcus, including type B; however, vaccination should not delay the start of eculizumab. Antibiotic prophylaxis is mandated during the first 2 weeks and is recommended for the overall treatment duration because not all serotypes are covered by vaccination.²⁵

It is not clear how long anti-C5 therapy should be extended, a relevant issue because of the very high cost of the drug. In addition, the risk for sensitization associated with chronic eculizumab exposure or with its deposition in tissue suggest that careful tapering to withdrawal whenever possible should be attempted, under tight control of disease and complement activity. In a prospective multicenter study of eculizumab discontinuation in 55 children and adults with aHUS, 13 patients experienced a relapse and this was predicted by the presence of a rare complement gene variant.⁴³ Different clinical courses before eculizumab therapy, and different residual complement activity while on eculizumab therapy, should be taken into consideration when strategies of chronic eculizumab therapy or discontinuation are planned. Reliable biomarkers of early relapse are strongly needed. In this regard, abnormally elevated serum-induced C5b-9 deposition on cultured endothelial cells highlighted aHUS relapses during eculizumab tapering/discontinuation.²⁷ At present, however, this test is available in only very few specialized centers.

Liver-kidney transplant. In patients with CFH mutations, combined liver-kidney transplant is aimed at correcting the genetic complement defect, thus preventing disease recurrence in the transplanted kidney (see Table 30.2). Liver transplantation cures aHUS without the need for specific therapies other than standard immunosuppression to prevent graft rejection. The short-term mortality risk associated with acute complement activation in the liver graft observed in initial attempts has been substantially reduced with prophylactic plasma exchange and perioperative eculizumab. Over 80% of patients with aHUS who received liver transplants to date with the previously described preparative regimen have had excellent long-term outcomes.^{44,45} However, the risks of kidney and liver transplantation have limited the widespread dissemination of this option.

SECONDARY HEMOLYTIC UREMIC SYNDROME

See Fig. 30.1 and Table 30.1.

Streptococcus pneumoniae–Associated HUS

Streptococcus pneumoniae accounts for 5% to 20% of reported pediatric HUS cases not associated with STEC.⁴⁶ Neuraminidase produced by *S. pneumoniae* cleaves N-acetylneuraminic acid from the glycoproteins on the cell membrane of erythrocytes, platelets, and glomerular cells; this exposes the normally hidden Thomsen-Friedenreich antigen (T antigen), which can then react with anti-T IgM antibodies naturally present in human serum. This reaction occurs more frequently in infants and children and causes polyagglutination of RBCs in vitro, so the Coombs test is often—although not always—positive, unlike in other forms of HUS. T-antigen exposure on RBCs is detected using the lectin *Hypogaea*. However, T-antigen exposure alone does not seem to be sufficient in causing HUS. Desialylation of cells also triggers complement activation via the alternative pathway. Severe consumption of complement components in serum during the acute phase and the identification of complement-related gene mutations in a few patients suggest a role for complement dysregulation in this form of HUS.⁴⁶

Patients, usually younger than 2 years, present with severe illness, including respiratory distress, neurologic involvement, and coma. The acute mortality rate is about 25%. The outcome hinges on the effectiveness of antibiotic therapy. In theory, plasma infusion or exchange is contraindicated because adult plasma contains antibodies against the T antigen that may accelerate polyagglutination and hemolysis. Thus, patients should be treated with only antibiotics, washed RBCs, and fluid and kidney replacement therapy. In some cases, plasma therapy with or without steroids has been associated with recovery.

Pregnancy-Associated HUS

HUS associated with pregnancy tends to occur at term or postpartum, within 3 months of delivery in most cases. This form is associated with genetic complement abnormalities in 50% to 60% of cases⁴⁷ and massive ex vivo serum-induced C5b-9 formation on endothelium⁴⁸ and should be included in the primary aHUS group, with pregnancy acting as a trigger on a predisposed genetic background. Consistently, around 90% of published patients with pregnancy-associated HUS underwent remission with eculizumab and the drug can be given safely during pregnancy.⁴⁹

HUS and Malignant Hypertension

Complement activation should also be suspected in patients with HUS in the setting of malignant hypertension.²⁶ Variants in complement genes have been reported in 35% to 65% of patients, and ex vivo serum-induced C5b-9 deposition on endothelial cells was increased, whereas it was normal in patients with malignant hypertension alone.^{27,50}

Blood pressure control is fundamental. Good response to eculizumab has been reported in a small series of patients. Early detection of complement activation via the endothelial assay or by elevation of plasma sC5b-9 is crucial to identify patients who would benefit from complement inhibition, especially when the disease does not improve with blood pressure control.

Posttransplant HUS

De novo posttransplant HUS has been reported in patients receiving kidney transplants or other solid organs and has been related to calcineurin inhibitors or humoral rejection. It occurs in 5% to 10% of kidney transplant patients receiving cyclosporine and in approximately 1% of those on tacrolimus. Withdrawal of the immunosuppressive agents is usually the first line of treatment. Underlying abnormalities in complement genes have been reported in one-third of cases. Anticomplement therapy with eculizumab is effective in some but not all cases.⁵¹ Positive C5b-9 staining in kidney biopsy and/or complement genetic abnormalities could be indications for eculizumab.

HUS complicates 10% to 40% of allogeneic bone marrow/hematopoietic stem cell transplants and is associated with high mortality.⁵² The disease is multifactorial. Risk factors include immunosuppressive drugs, graft versus host disease, chemotherapy, radiation, and infections. In a large study, around 60% of patients with HSCT and HUS had at least one complement gene variant compared with 9% of patients without HUS,⁵³ and plasma sC5b-9 levels were elevated in the former. Other authors reported a lower incidence of complement gene abnormalities (20%–30%). Treatment remains controversial. Favorable outcomes with eculizumab have been described in a few cases.

HUS Associated With Autoimmune and Systemic Diseases

Several autoimmune diseases can be associated with HUS, including systemic lupus erythematosus (SLE), antiphospholipid syndrome, and scleroderma. The pathophysiology of HUS in these diseases is multifactorial. Although genetic complement abnormalities are rare in these patients,⁵⁴ complement activation via the classical or the alternative pathways occurs secondary to the underlying autoimmune disease (see Fig. 30.1 and Table 30.1).

Treatment of the concomitant disease should be the first line of intervention. Plasma therapy should always be attempted, even though its efficacy is poorly defined. In a few patients with persistent HUS despite treatment of SLE, the use of eculizumab was associated with an improvement in hematologic indices and/or kidney function.^{26,55}

In the antiphospholipid syndrome (APS; see Chapter 29), oral anticoagulation remains the only treatment of proven efficacy to prevent and treat microvascular and macrovascular thrombosis. Rituximab may be efficacious, but further controlled studies are needed.⁵⁶ Less than 1% of patients with APS develop catastrophic APS (defined as intravascular thrombosis affecting three or more systems despite any treatments). Eculizumab has been used to treat these refractory cases, and published data showed improvement or stabilization.²⁶ Blood pressure control is fundamental in TMA associated with scleroderma crisis.

Drug-Induced HUS

Drug-induced HUS may result from direct toxicity of the drug to endothelial cells or is immune mediated (see Fig. 30.1 and Table 30.1). TMA, commonly resembling HUS, occurs in 2% to 10% of cancer patients treated with mitomycin C and usually in remission from their malignancy.⁵⁷ The fatality rate is about 70%; patients surviving the acute phase often remain on chronic dialysis or die later of recurrence of the tumor or metastases. Preventing the disease by corticosteroids during mitomycin treatment has been suggested and needs to be confirmed. Plasma exchange is usually attempted, but its effectiveness is unproven.

Quinine is one of the most common drugs associated with HUS.⁵⁷ It is generally used to treat muscle cramps but is also contained in beverages and nutrition health products. Quinine-dependent platelet, erythrocyte, granulocyte, lymphocyte, and endothelial antibodies may contribute to the pathogenesis. Severe kidney impairment is frequent, and hemodialysis is required in most cases. High rates of death and CKD have been reported. Quinine cessation and plasma therapy should be initiated rapidly.

TMA associated with interferon alfa is characterized by predominant kidney impairment. Recovery of the disease is common in cases of early discontinuation of the drug and prompt supportive therapy. However, kidney prognosis is usually poor, with ESKD reported in about 42% of cases.

Variants in complement genes are rare in drug-induced HUS (prevalence not different from controls) and the response after drug withdrawal was not different in patients treated with plasma, corticosteroids, or eculizumab.^{26,54}

Infection-Associated HUS

HUS can occur in infections by viral (cytomegalovirus [CMV], Epstein-Barr virus, varicella zoster virus, parvovirus, human immunodeficiency virus [HIV], West Nile virus), bacterial, and other agents (*Bordetella pertussis*, *Mycoplasma pneumoniae*, *Toxoplasma gondii*, *Plasmodium vivax*).²⁶ Mechanisms are complex and include direct endothelial injury and complement activation involving any of the three activation pathways. The incidence of alternative pathway genetic abnormalities is low and varies according to the type of infection.²⁶

In COVID-19 disease (see Chapter 59), there is increasing evidence that unrestrained complement activation induced by the virus in the lungs and other organs is one of the mechanisms of tissue injury, endothelial dysfunction, and TMA-like thrombosis.⁵⁸ Elevated levels of sC5b-9 and C5a have been reported in the plasma of severe COVID-19 patients, indicating terminal complement activation, and preliminary data of C5 or C5a inhibition in a few patients with severe COVID-19 are encouraging. More than 10 clinical trials of complement inhibition in COVID-19 are underway.⁵⁸

Malignancy

Spontaneous TMA complicates almost 6% of cases of metastatic gastric carcinoma, which, in turn, accounts for about 50% of all malignant

disease-associated TMA (see Fig. 30.1 and Table 30.1). The prognosis is extremely poor. Therapy is minimally effective.

Cobalamin C Deficiency–Associated HUS

HUS can manifest in methylmalonic aciduria and homocystinuria caused by recessive mutations in the *MMACHC* gene, which result in deficiency of cobalamin C type (cblC), the most common genetic functional variant of cobalamin (vitamin B₁₂) (see Fig. 30.1 and Table 30.1). It is a rare condition, and around 40 cases of HUS associated with cblC defect have been reported. Pulmonary hypertension may accompany the disease. Diagnosis is facilitated by finding elevated plasma homocysteine and methylmalonic aciduria. HUS usually presents in early infancy, although adult onset has also been reported. Mortality is high if untreated, but metabolic therapy with hydroxocobalamin is very effective.

In summary, the term secondary HUS (or secondary TMA) includes a very heterogeneous group of conditions. Complement activity tests, like the serum-induced C5b-9 formation on endothelium, and genetic screening could allow us to reclassify each patient on etiology basis and determine whether and to what extent complement activation plays a role in the disease. Massive ex vivo C5b-9 formation on endothelium during acute episodes²⁷ indicates complement as a driving factor and may be used to select patients for complement inhibition. Those patients also carrying genetic complement abnormalities and/or showing chronically increased C5b-9 formation in remission²⁷ are most likely primary aHUS cases genetically predisposed to complement dysregulation, in whom the coexisting condition acted as a trigger of the acute episode (see Fig. 30.1 and Table 30.1).

Thrombotic Thrombocytopenic Purpura

In the microvasculature of patients with TTP, systemic platelet thrombi develop, mainly formed by platelets and von Willebrand factor (vWF). vWF plays a major role in primary hemostasis, forming platelet plugs at sites of vascular injury under high shear stress. vWF is synthesized in vascular endothelial cells and megakaryocytes as a high-molecular-weight polymer and is stored in the Weibel-Palade bodies. On stimulation, vWF is secreted by endothelial cells as ultra-large (UL) multimers that form string-like structures attached to the endothelial cells, possibly through interaction with P-selectin. Under fluid shear stress, the UL-vWF strings are cleaved to generate the range of vWF multimer sizes that normally circulate in the blood.⁵⁹ The proteolytic cleavage of vWF multimers appears to be critical to prevent thrombosis in the microvasculature.

Liver-derived ADAMTS13 is the protease responsible for cleaving vWF, creating 140- and 176-kDa fragments (Fig. 30.9). ADAMTS13 is also expressed at lower levels in renal podocytes and tubular cells, vascular endothelial cells, and platelets. The plasma concentration of ADAMTS13 is approximately 1 µg/mL (5 nmol/L). The elimination half-life of ADAMTS13 is 1 to 2 days in the circulation.

Mechanisms

ADAMTS13 is severely deficient in patients with primary TTP, leading to accumulation of UL-vWF multimers that are highly reactive with platelets.⁶⁰ Exposure of UL-vWF multimers and platelets to shear stress leads to platelet aggregation. vWF and platelets have a propensity to form aggregates at the levels of shear stress found in normal arterioles and capillaries, and this needs to be constantly regulated. By cleaving UL-vWF multimers before they are activated by shear stress to cause platelet aggregation, ADAMTS13 prevents spontaneous microvascular thrombosis in the normal circulation (see Fig. 30.9). The platelet aggregation observed in patients with TTP and ADAMTS13 deficiency is thus a direct consequence of the accumulation of UL-vWF

multimers. Consequently, microvascular thrombi occur in almost all organ vessels, resulting in diffuse organ ischemia and thrombocytopenia secondary to platelet consumption.

Two mechanisms for deficiency of the ADAMTS13 activity include an acquired deficiency secondary to the formation of anti-ADAMTS13 autoantibodies (acquired TTP) and a genetic deficiency resulting from homozygous or compound heterozygous mutations in *ADAMTS13* (genetic TTP; see Fig. 30.1).

TTP associated with Anti-ADAMTS13 antibodies. This form accounts for the majority of acute cases (60%–90%) and is characterized by severe functional deficiency of ADAMTS13 because of transient, specific autoantibodies that tend to disappear during remission, and this is paralleled by normalization of ADAMTS13 activity (see Fig. 30.1 and Table 30.1).⁶⁰ Patients with TTP secondary to malignancies or HIV infection rarely have severe ADAMTS13 deficiency and inhibitory IgG antibodies.⁶¹ Severe ADAMTS13 deficiency and ADAMTS13 inhibitory antibodies were detected in 80% to 90% of patients with ticlopidine-associated TTP and in a few patients with clopidogrel-induced TTP (see Table 30.1). The deficiency resolved after the drugs were discontinued. TTP diagnosed during pregnancy—most frequently during the second and third trimesters—accounts for 10% to 30% of all adult cases of TTP.⁴⁹ Most of these cases have acquired ADAMTS13 deficiency, but pregnancy also has been reported as a triggering event in patients with genetic ADAMTS13 deficiency (see later discussion).

TTP associated with genetic deficiency of ADAMTS13. This shiftenterrare inherited form of TTP is associated with homozygous or compound heterozygous mutation in the *ADAMTS13* gene and accounts for about 5% of all of cases of TTP (see Table 30.1). The disease is inherited as a recessive trait in patients with and without a family history of TTP⁶²; more than 120 *ADAMTS13* mutations have been identified. Studies on secretion and activity of the mutated forms of *ADAMTS13* showed that most of these mutations led to impaired secretion of the protease from the cells, and when the mutated protein is secreted, the proteolytic activity is greatly reduced.

Most patients are carriers of compound heterozygous mutations; only 20% of mutations have been observed in homozygous form.

Clinical Course

Compared with patients with less severe ADAMTS13 deficiency, severely deficient patients experience a higher proportion of therapy-induced remissions (82%–88% vs. 20%–75%) and lower mortality (8%–18% vs. 18%–80%).⁶¹ The high mortality risk in non-severely deficient patients may be because of the higher proportion of secondary causes and death from underlying diseases, such as patients with hematologic malignancies.

Among patients who have a severe ADAMTS13 deficiency, those with inhibitory antibodies experience more severe disease, take a substantially longer time to achieve clinical remission, and require a higher plasma volume than patients with genetic ADAMTS13 deficiency. In patients with inhibitory antibodies, up to 50% develop relapses and undetectable ADAMTS13 activity and persistence of anti-ADAMTS13 inhibitors during remission predict recurrences.

TTP has been reported in 1 in every 1600 to 5000 patients treated with ticlopidine, and 11 cases have been reported during treatment with clopidogrel. The overall survival rate is 67% and is improved by early treatment withdrawal and plasma therapy.

Approximately 60% of patients with genetic deficiency of ADAMTS13 experience their first acute TTP episode in the neonatal period or during infancy. A second group (10%–20%) manifests the disease after the third decade of life. TTP relapses occur in 80% of patients, but their frequency varies widely. Whereas some patients

Pathophysiology of Platelet Aggregation in Thrombotic Thrombocytopenic Purpura

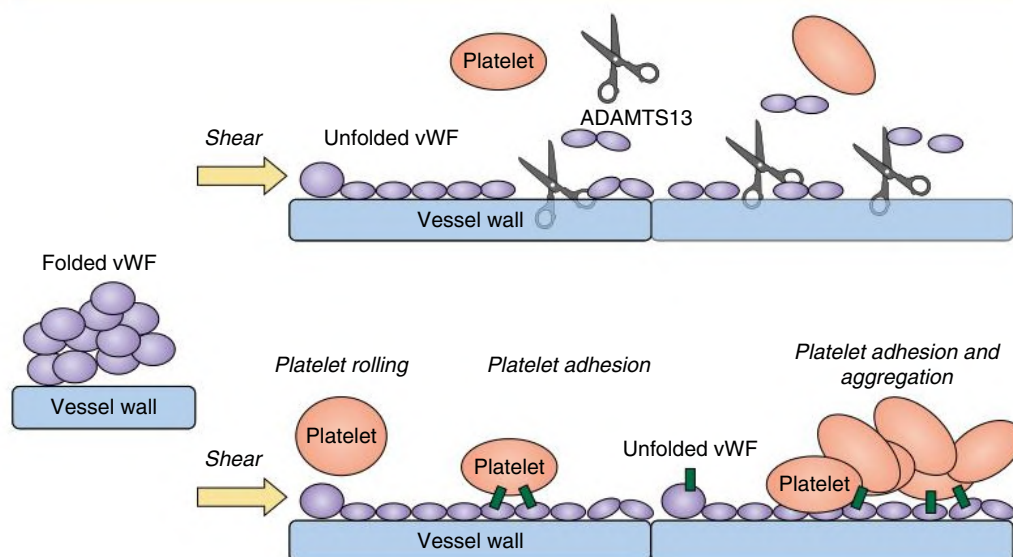


Fig. 30.9 The Pathophysiology of Platelet Aggregation in Thrombotic Thrombocytopenic Purpura. Von Willebrand factor (vWF) is synthesized and stored as ultra-large (UL) multimers in endothelial cells and megakaryocytes. On stimulation, UL-vWF multimers are secreted by endothelial cells into the circulation in a folded structure. On exposure to enhanced shear stress, UL multimers form string-like structures that adhere to endothelial cells. Normally, UL-vWF strings are cleaved by ADAMTS13 to generate vWF multimers from 500 kDa to 20 million Da and prevent thrombosis in the microvasculature (*upper panel*). When the ADAMTS13 proteolytic activity is defective because of the inhibitory effect of anti-ADAMTS13 autoantibodies or congenital defective synthesis of the protease, UL-vWF multimers accumulate and interact with activated platelets to facilitate platelet adhesion and aggregation, with thrombi formation and occlusion of the vascular lumen (*lower panel*).

with genetic ADAMTS13 deficiency depend on frequent chronic plasma infusions to prevent relapses, others remain free of disease for long periods after plasma discontinuation. The type and location of *ADAMTS13* mutations may influence the age of onset of TTP and the penetrance of the disease in mutation carriers. Mutations resulting in very low residual ADAMTS13 activity (<1%) were associated with childhood onset, relapsing disease, and kidney impairment.⁸

Environmental factors may contribute to induce full-blown manifestation of the disease. According to this two-hit model, deficiency of ADAMTS13 predisposes to microvascular thrombosis and TMA supervenes after a triggering event that activates microvascular endothelial cells and causes the secretion of UL-vWF multimers. Potential triggers of these phenomena are infections and pregnancy.

Therapy

In patients with acquired TTP, plasma is the cornerstone of therapy in an acute episode because it replaces defective protease activity. Plasma exchange, compared with infusion, also rapidly removes anti-ADAMTS13 antibodies (see [Table 30.2](#)).

Because of the potential for sudden clinical deterioration, treatment should be initiated as soon as possible after diagnosis. Treatment consists of a daily 1 to 2 plasma volume exchanges until clinical symptoms have resolved and the platelet count is stably at least 150,000/ μ L. Fresh-frozen plasma and cryosupernatant plasma are considered equivalent because of comparable levels of ADAMTS13.

Corticosteroids (see [Table 30.2](#)) given in combination with plasma therapy may benefit autoimmune forms of TTP by inhibiting the synthesis of anti-ADAMTS13 autoantibodies; plasma

exchange will have only a temporary effect on the autoimmune basis of the disease. Two randomized clinical trials (phase 2 TITAN study and phase 3 HERCULES trial) in patients with acquired TTP found that caplacizumab, an anti-vWF humanized single variable domain immunoglobulin (nanobody), combined with plasma exchange and immunosuppressive therapy (see later), was associated with a shorter time to normalization of the platelet count than plasma exchange and immunosuppression plus placebo.⁶³ Caplacizumab also improved thromboembolic event rate, death rate, and relapse rate in acquired TTP.⁶⁴ Caplacizumab reduces microvascular thrombosis by inhibiting interaction between vWF multimers and platelets, but it does not affect the production of anti-ADAMTS13 autoantibodies. Thus, plasma exchange and immunosuppressive treatment are still an essential part of TTP treatment.

Prospective studies have used rituximab (see [Table 30.2](#)) in patients who failed to respond to standard daily plasma exchange and methylprednisolone and in patients with relapsed acute TTP who previously had antibodies to ADAMTS13. Treatment was associated with clinical remission, disappearance of anti-ADAMTS13 antibodies, and increase of ADAMTS13 activity to levels greater than 10%. Time to remission has been variable, from 1 to 4 weeks after the first dose. The duration of remission has ranged between 9 months and 4 years, with relapses reported in approximately 10%. Rituximab also has been used electively to prevent relapses in patients with autoantibodies.⁶⁵ Longitudinal evaluation of ADAMTS13 activity and autoantibody levels may help monitor patient responses to treatment. Retreatment with rituximab should be considered when ADAMTS13 activity decreases and inhibitors reappear into the circulation to prevent a relapse.

In genetic forms of TTP, ADAMTS13 is constitutively lacking and can be replaced by plasma therapy. During acute episodes, patients often require plasma exchange. Providing enough ADAMTS13 to achieve 5% normal enzymatic activity may be sufficient to degrade large vWF multimers—which may induce remission of the microangiopathic process—and this effect is sustained over time. Infused ADAMTS13 has a plasma half-life of 2 to 3 days in vivo, and although plasma levels fall below 5% within 3 to 7 days after plasma administration, the effect of plasma on platelet counts and clinical parameters may last up to 3 weeks, suggesting that ADAMTS13 remains

available, for example, on platelets and endothelial cells. Patients with genetic ADAMTS13 deficiency tend to relapse. Patients with frequent relapses, a severe clinical course with neurologic sequelae, kidney insufficiency, and siblings who have died of TTP should be put on regular prophylactic plasma infusions every 2 to 3 weeks, which prevents episodes of acute TTP and maintains the patients in good health for years. Recombinant ADAMTS13 has demonstrated efficacy in increasing ADAMTS13 activity in a phase I/II study in patients with genetic TTP. A phase III clinical trial is ongoing (NCT03393975).

SELF-ASSESSMENT QUESTIONS

- Which are the common clinical features characterizing patients with thrombotic microangiopathies?
 - Thrombocytopenia and nonimmune hemolytic anemia, with or without neurologic and/or renal dysfunction
 - Diarrhea and kidney dysfunction
 - Anemia, antiplatelet antibodies, and purpura
 - Disseminated intravascular coagulation
- Which is the *most* common form of thrombotic microangiopathy?
 - Atypical hemolytic uremic syndrome
 - Shiga toxin *E. coli*-associated hemolytic uremic syndrome
 - Thrombotic thrombocytopenic purpura
 - Antiphospholipid syndrome
- Which is the *most* common genetic abnormality associated with aHUS?
 - C3 mutations
 - ADAMT13 mutations
 - CFH mutations
 - CFTR mutations
- Which of the following is *not* indicated for treatment of aHUS?
 - Conservative therapy
 - Eculizumab
 - Plasma exchange
 - Antibiotics
- For which of the following is there an indication for rituximab?
 - TTP with congenital ADAMTS13 deficiency
 - Shiga-toxin producing *Escherichia coli*-associated HUS
 - TTP with anti-ADAMTS13 antibodies
 - aHUS associated with MCP mutations

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